2.1.1 ▶ The Maxwell-Boltzmann Distribution

An ideal gas can be represented by a Gaussian distribution of the velocity of the singular particles in the gas, given by the following distribution:

$$f(v) = \frac{4}{\sqrt{\pi}} \left[\frac{m}{2k_B T}\right]^{3/2} v^2 e^{\frac{-mv^2}{2k_B T}} \tag{2.1}$$

The function is known as the **Maxwell-Boltzmann Distribution** and although it is a bit messy, it's important to recognize what it means.

The function is a **Probability Distribution Function (PDF)**, which is a special type of function that describes the way in which a random system is arranged. If we consider an arbitrary PDF $\rho(x)$ where x is some state of the system, ρ has no real meaning at any value of x. However, the area under ρ (integral) represents the probability of the sytem taking on a state that lies between the bounds of integration. Thus, any PDF $\rho(x)$ has the property

$$\int_{-\infty}^{\infty} \rho(x)dx = 1 \tag{2.2}$$

since the system must always be occupying one of its possible states. Similarly, the probability of an event occurring in the interval $(x, x + dx)$ is $\rho(x)dx$. For a container of gas, the particles are all evenly distributed, meaning that the probability of a gas particles having a velocity in the range $(v, v + dv)$ equals the fraction of all the particles with that velocity. Or,

$$\frac{dN}{N} = f(v)dv \tag{2.3}$$

where dN is the number of gas particles with a speed in the range $(v, v + dv)$. Recall that the expected value (or mean) of a system is defined as

$$\langle x \rangle = \sum_i x_i p(x_i)$$

where $\langle x \rangle$ is the average of x, x_i is an arbitrary state, and $p(x_i)$ is the probability of x_i occurring. It follows that for continuous distributions

$$\langle x \rangle = \int_{-\infty}^{\infty} x\rho(x)dx \tag{2.4}$$

since x is a state, $\rho(x)dx$ is the probability, and we are again summing over all possible states. Similarly, the mean of any function of x is

$$\langle g(x) \rangle = \int_{-\infty}^{\infty} g(x)\rho(x)dx.^{1} \tag{2.5}$$

[1] The reason why we don't change $\rho(x)$ to say, $g(\rho(x))$ is because the fundamental randomization process remains unchanged, and we are simply rescaling the quantity. For example, if we wanted to find the average of the square of a die roll, we would multiply the outcomes $(1, 4, 9, 16, 25, 36)$ by the probability $1/6$ and add them up to find the mean, which is the same probability used to calculate the mean value of the die normally. This all makes sense, because we aren't changing anything about the die roll, just the output value of each state. At any rate, these properties will be important when we examine macro-scale properties of systems of small particles.

Going back to the Maxwell-Boltzmann distribution, we can break it into two parts:

$$f(v) = \frac{4}{\sqrt{\pi}} \left[\frac{m}{2k_B T} \right]^{3/2}$$
$$\cdot\, v^2 e^{\frac{-mv^2}{2k_B T}}$$

The bottom part of the distribution relates to the actual distribution of the speeds of the particles in the gas[2] and is derived using statistical physics and experiment. The top part is a normalization factor, which gives the function the property from eq. 2.2. The Maxwell-Boltzmann distribution is special, though, since it's normalized in such a way that the area under the curve from 0 to ∞ is one, instead of the area along the entire x axis. To see why, consider the following graph of $f(v)$:

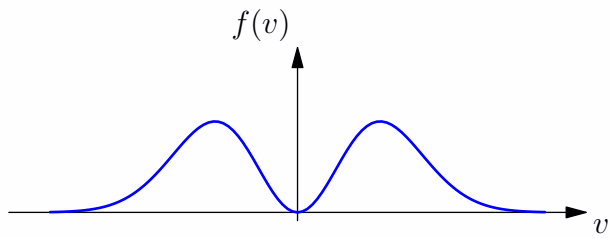

Fig. 2.1: Maxwell-Boltzmann Distribution

Since objects cannot have negative speed, all of the particles in a gas must lie on the right side of the graph, meaning

$$\int_0^\infty f(v)dv = 1. \tag{2.6}$$

This somewhat unusual normalization will be important in 2.1.2 when we integrate over all particles ★

[2] i.e. it's the part that gives the graph its shape